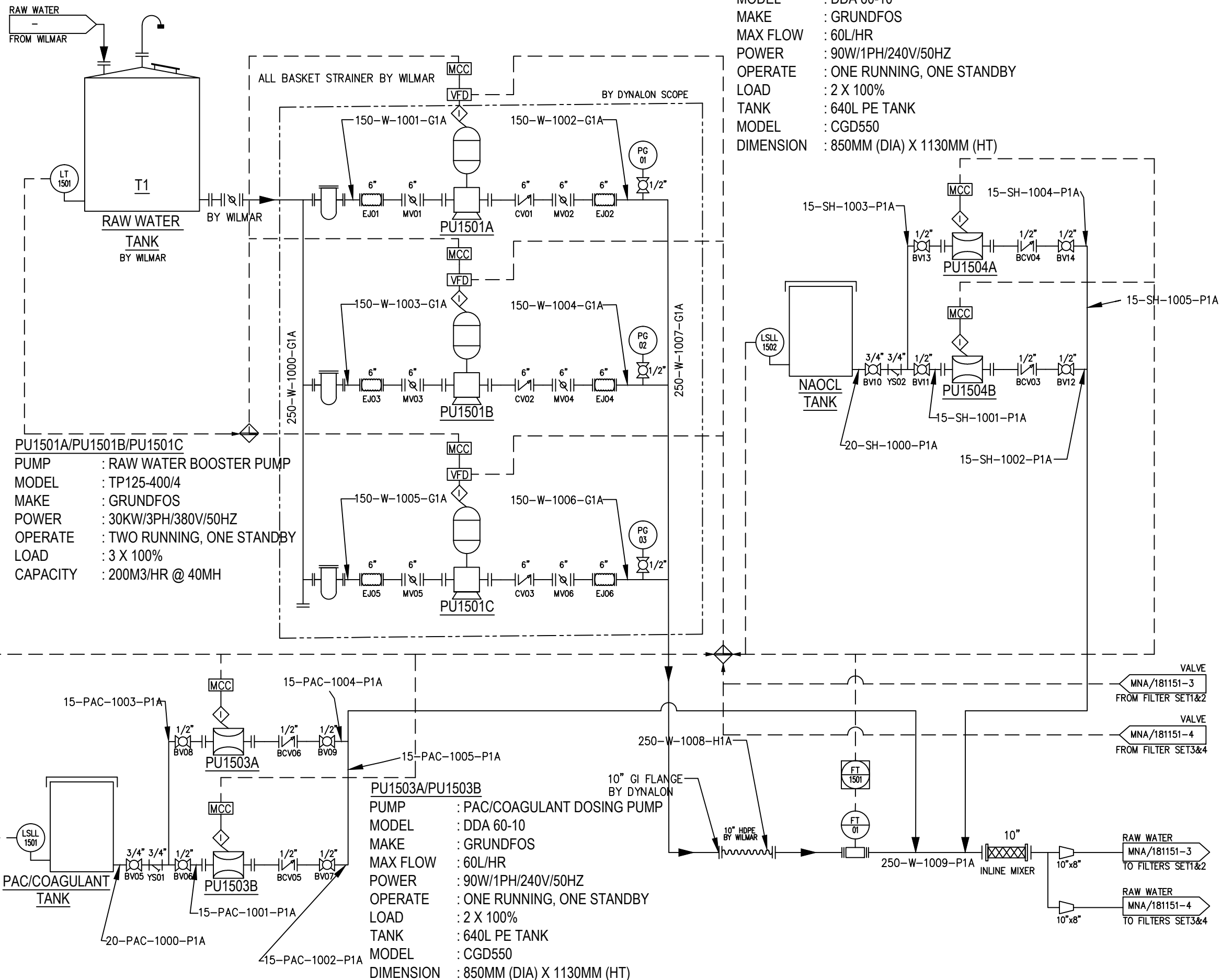




PU1504A/PU1504B
PUMP : NAOCL DOSING PUMP
MODEL : DDA 60-10
MAKE : GRUNDFOS
MAX FLOW : 60L/HR
POWER : 90W/1PH/240V/50HZ
OPERATE : ONE RUNNING, ONE STANDBY
LOAD : 2 X 100%
TANK : 640L PE TANK
MODEL : CGD550
DIMENSION : 850MM (DIA) X 1130MM (HT)

PU1501A/PU1501B/PU1501C
PUMP : RAW WATER BOOSTER PUMP
MODEL : TP125-400/4
MAKE : GRUNDFOS
POWER : 30KW/3PH/380V/50HZ
OPERATE : TWO RUNNING, ONE STANDBY
LOAD : 3 X 100%
CAPACITY : 200M3/HR @ 40MH

PU1503A/PU1503B
PUMP : PAC/COAGULANT DOSING PUMP
MODEL : DDA 60-10
MAKE : GRUNDFOS
MAX FLOW : 60L/HR
POWER : 90W/1PH/240V/50HZ
OPERATE : ONE RUNNING, ONE STANDBY
LOAD : 2 X 100%
TANK : 640L PE TANK
MODEL : CGD550
DIMENSION : 850MM (DIA) X 1130MM (HT)

LEGEND

REV-17	14/11/2018	CHANGE TAG NO.
REV-16	29/10/2018	ADMEND MMF INLET PIPE & MIXER TO 10", ADD CR
REV	DATE	REMARKS

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
2X135M3/HR UF & 2X100M3/HR
RO SYSTEM

DRAWING TITLE :

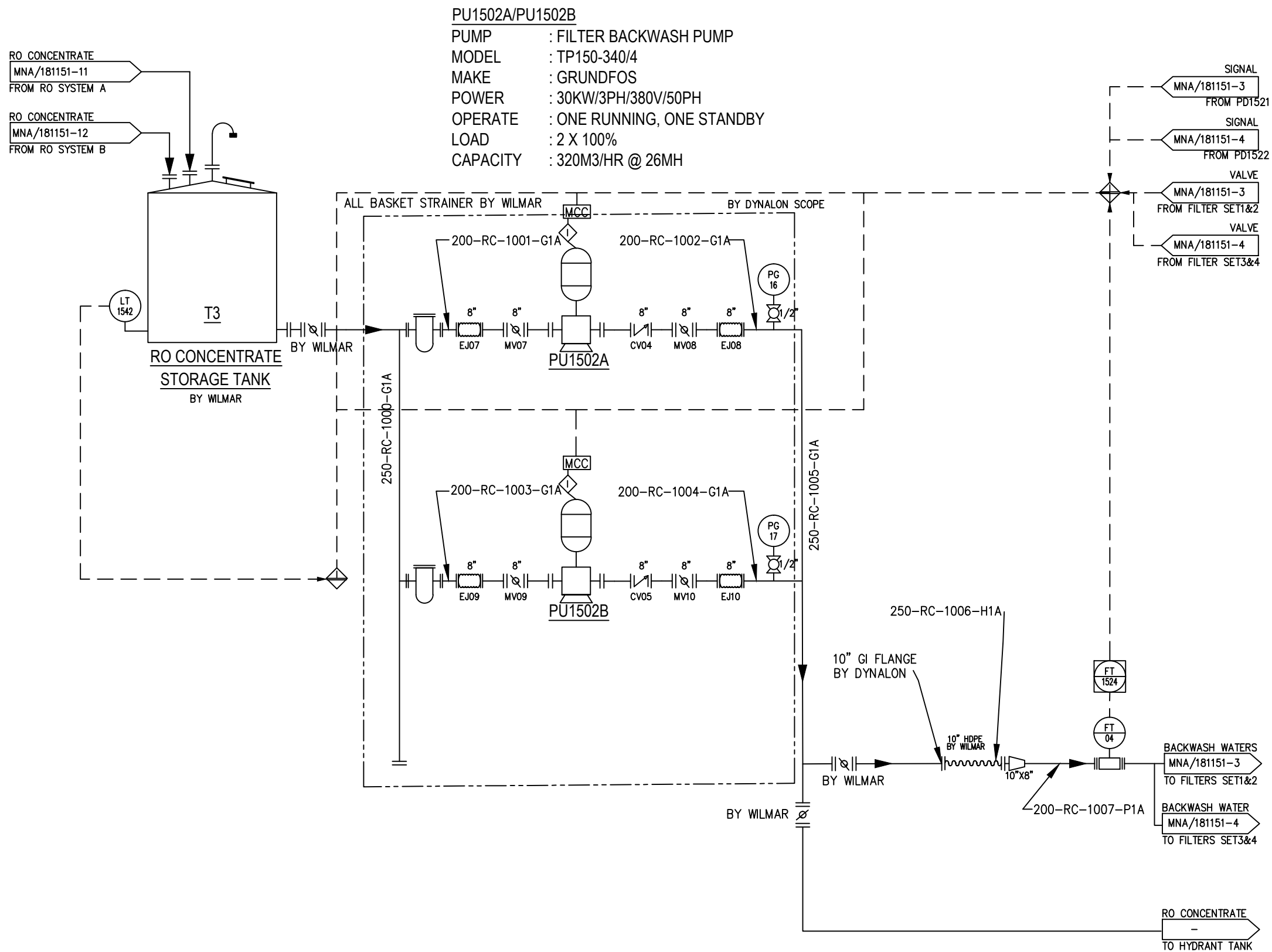
PROCESS INSTRUMENTATION DIAGRAM

CONTRACTOR :

DYNALON ENGINEERING SDN BHD
NO 5, JALAN P2/19A, BANDAR TEKNOLOGI KAJANG,
43500 SEMENYIH SELANGOR, MALAYSIA.
TEL : +603-87236991 / 7991 FAX : +603-87248991

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DESIGNED BY	CH SHO	APPROVED BY	
DRAWN BY	JUNE	SCALE	NTS
DATE	19.07.2018	DRAWING NO.	MNA/181151-1 REV-17



LEGEND

REV	DATE	REMARKS
REV-17	14/11/2018	CHANGE TAG NO.

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
2X135M3/HR UF & 2X100M3/HR
RO SYSTEM

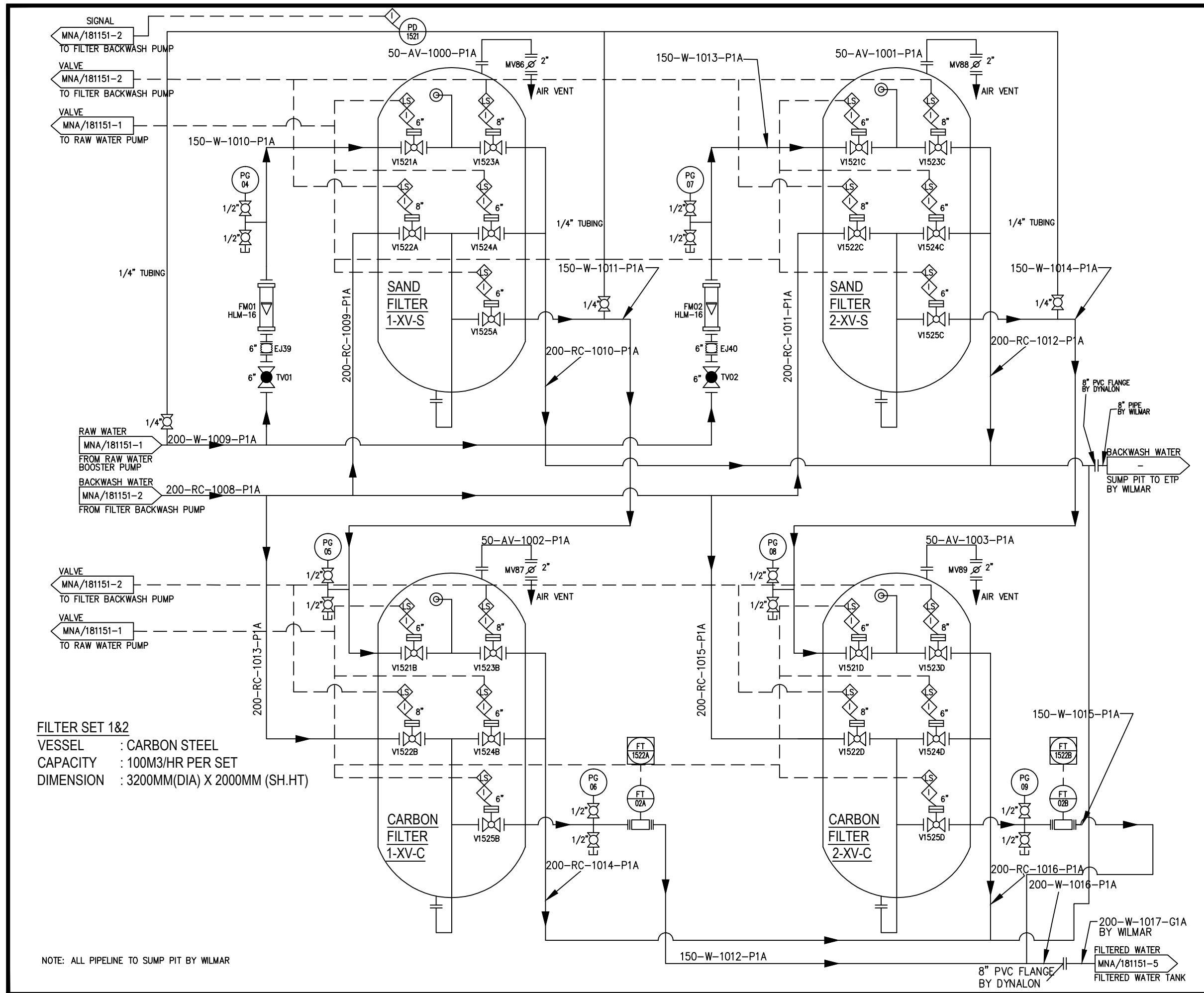
DRAWING TITLE :

PROCESS INSTRUMENTATION DIAGRAM

CONTRACTOR :
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DATE	19.07.2018	DRAWING NO.	MNA/181151-2 REV-17



LEGEND

REV	DATE	REMARKS
REV-17	14/11/2018	CHANGE TAG NO.
REV-16	29/10/2018	ADMEND MMF INLET PIPE TO 10". ADDED CR.

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
2X135M3/HR UF & 2X100M3/HR
RO SYSTEM

DRAWING TITLE :

PROCESS INSTRUMENTATION DIAGRAM

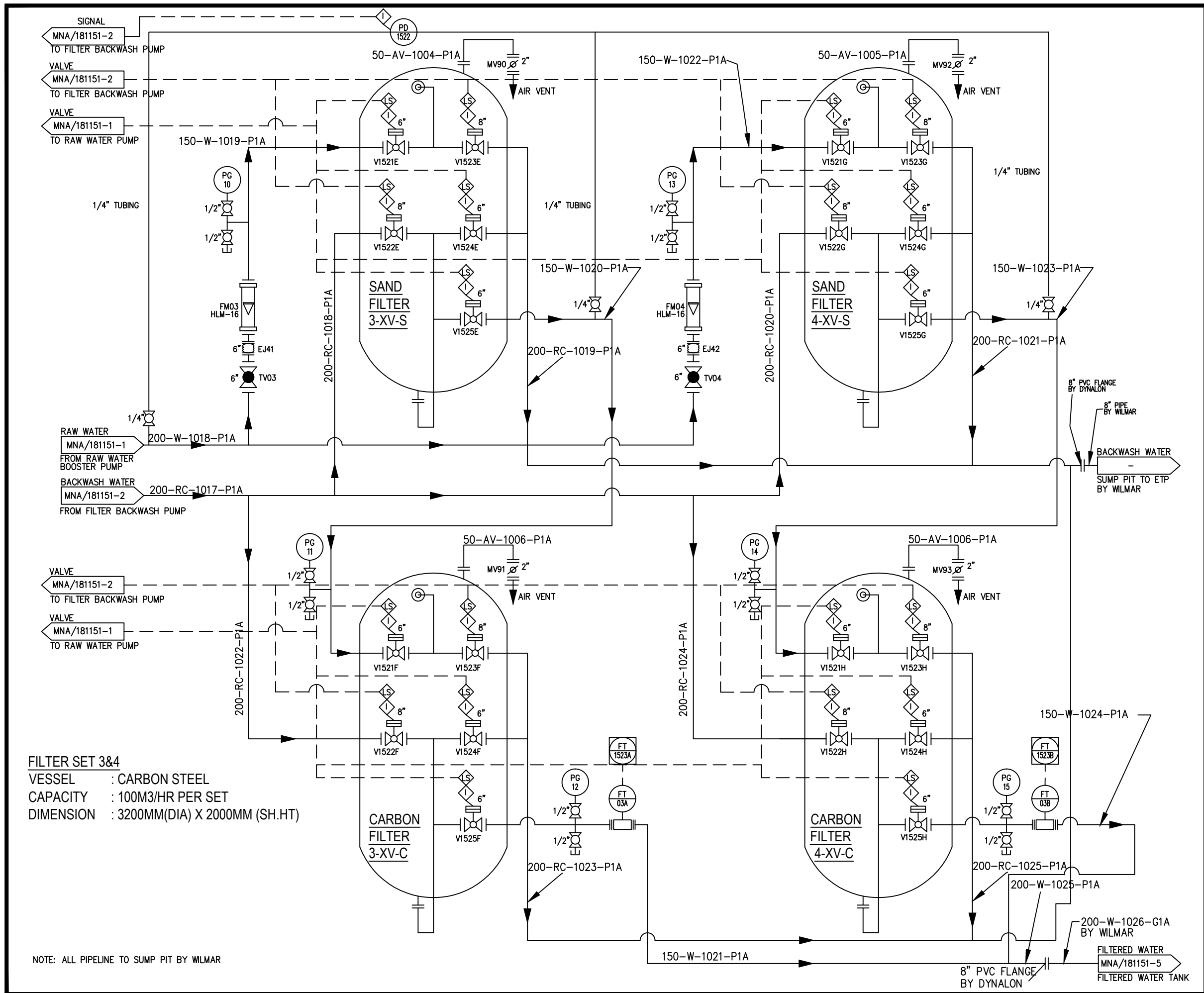
CONTRACTOR :



DYNALON ENGINEERING SDN BHD
NO 5, JALAN P2/19A, BANDAR TEKNOLOGI KAJANG,
43500 SEMENYIH SELANGOR, MALAYSIA.
TEL : +603-87236991 / 7991 FAX : +603-87248991

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DESIGNED BY	CH SHO	APPROVED BY	
DRAWN BY	JUNE	SCALE	NTS
DATE	19.07.2018	DRAWING NO.	MNA/181151-3 REV-17



FILTER SET 3&4
 VESSEL : CARBON STEEL
 CAPACITY : 100M3/HR PER SET
 DIMENSION : 3200MM(DIA) X 2000MM (SH.HT)

LEGEND

REV-17 14/11/2018 CHANGE TAG NO.

REV DATE REMARKS

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
 2X135M3/HR UF & 2X100M3/HR
 RO SYSTEM

DRAWING TITLE :

PROCESS INSTRUMENTATION DIAGRAM

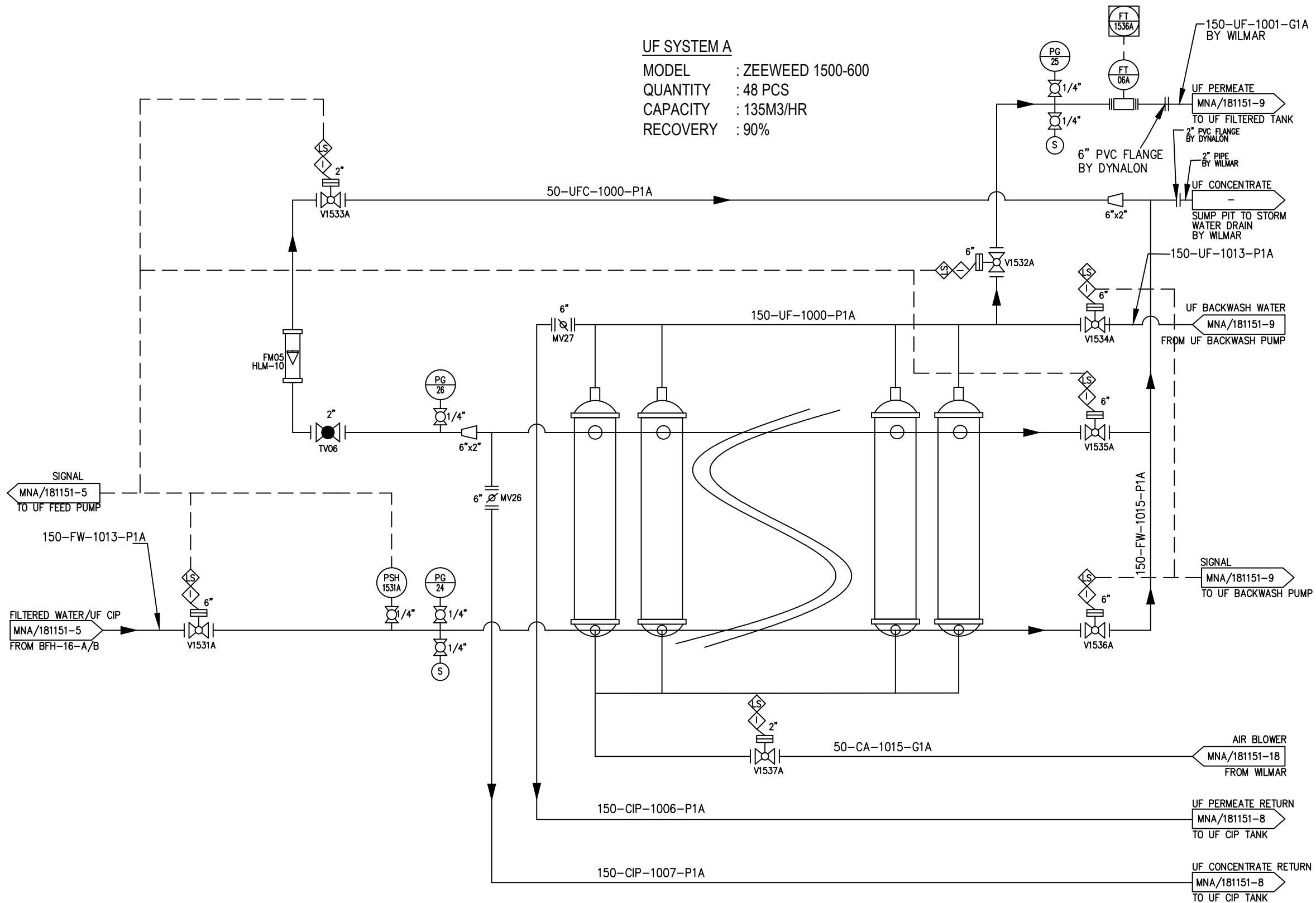
CONTRACTOR :



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 43500 SEMENYIH SELANGOR, MALAYSIA.
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DATE	19.07.2018	DRAWING NO.	MNA/181151-4 REV-17



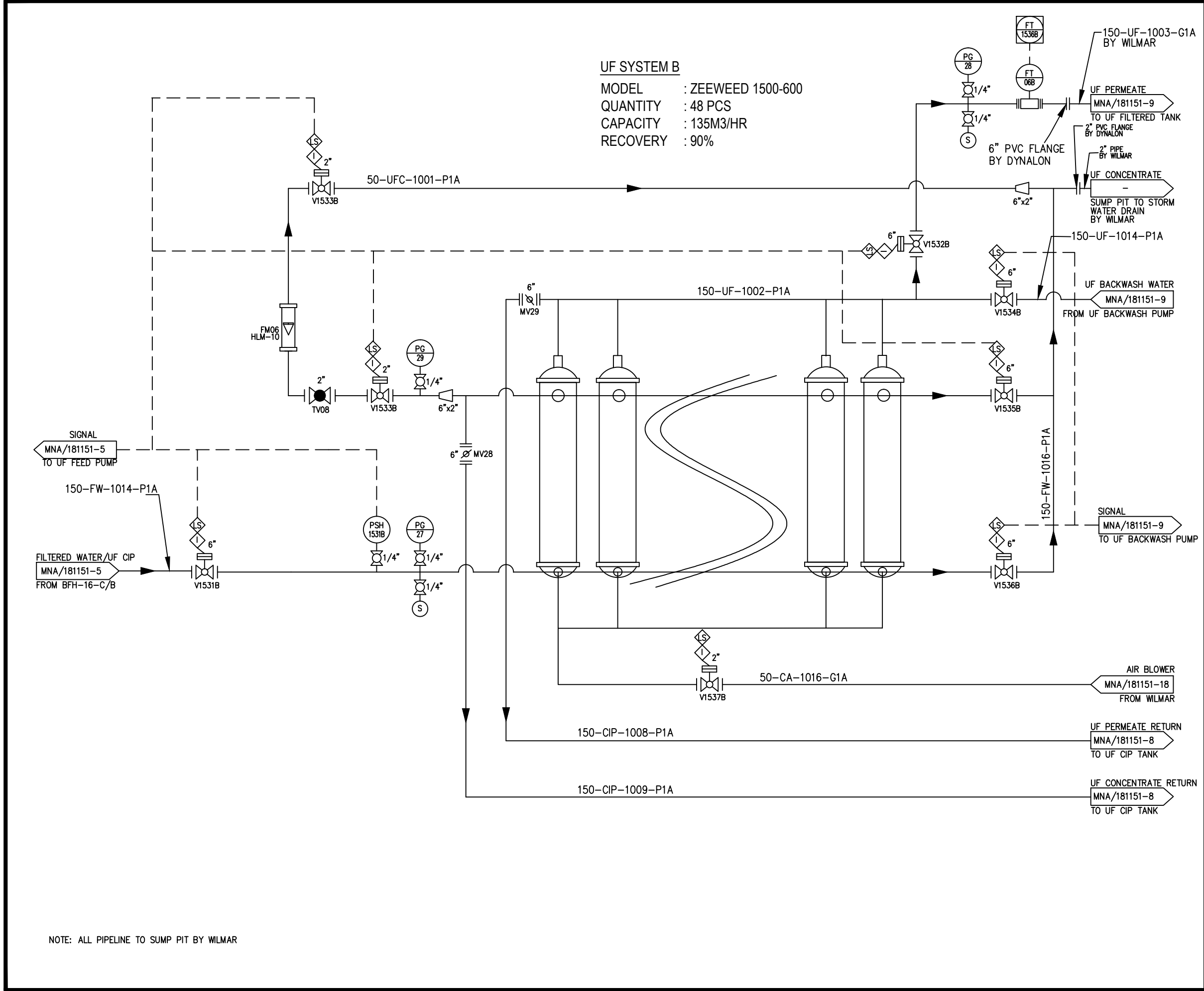
LEGEND

REV-17	14/11/2018	CHANGE TAG NO. / RELOCATE FT1535A
REV-17	14/11/2018	RELOCATE V1533A
REV	DATE	REMARKS

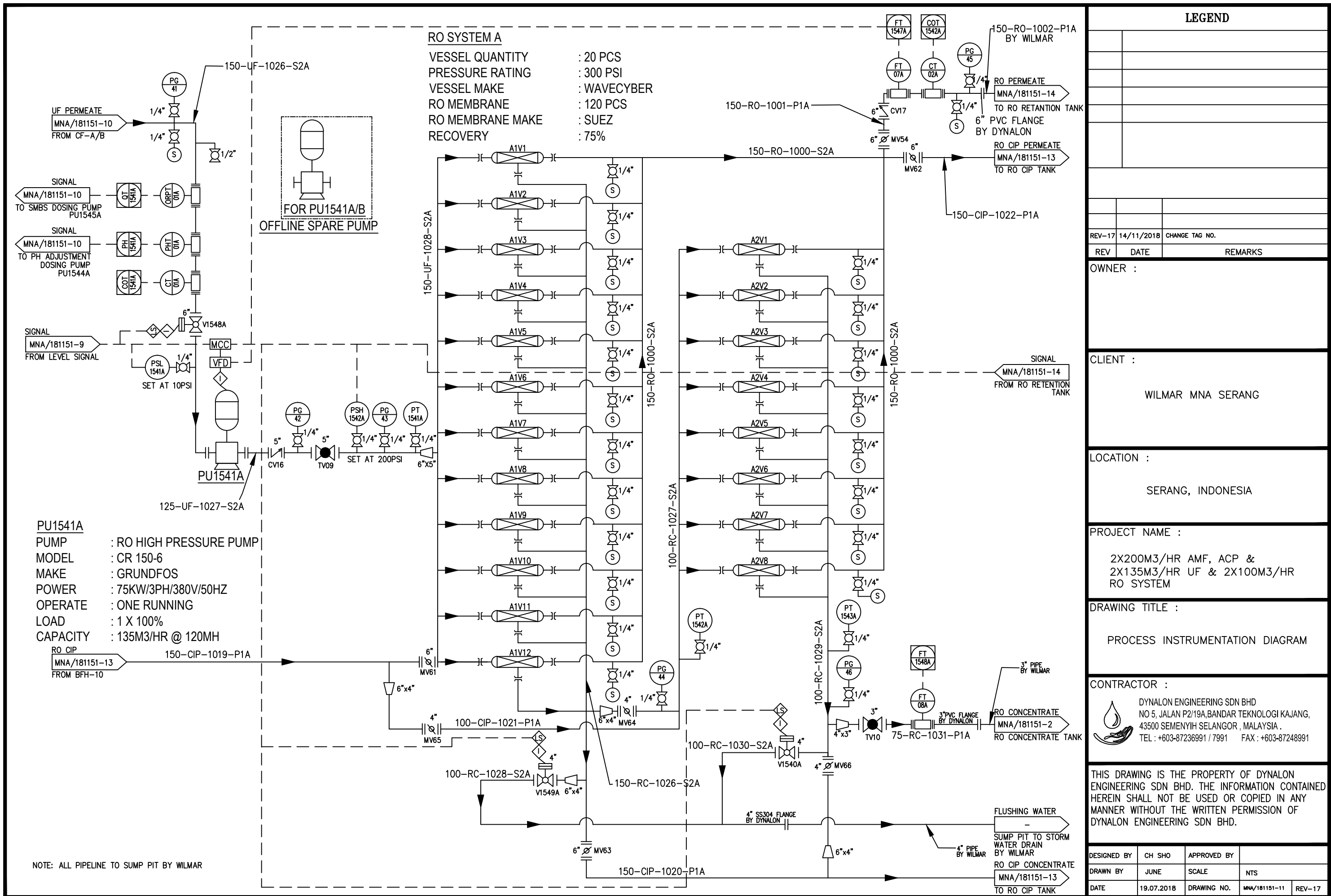


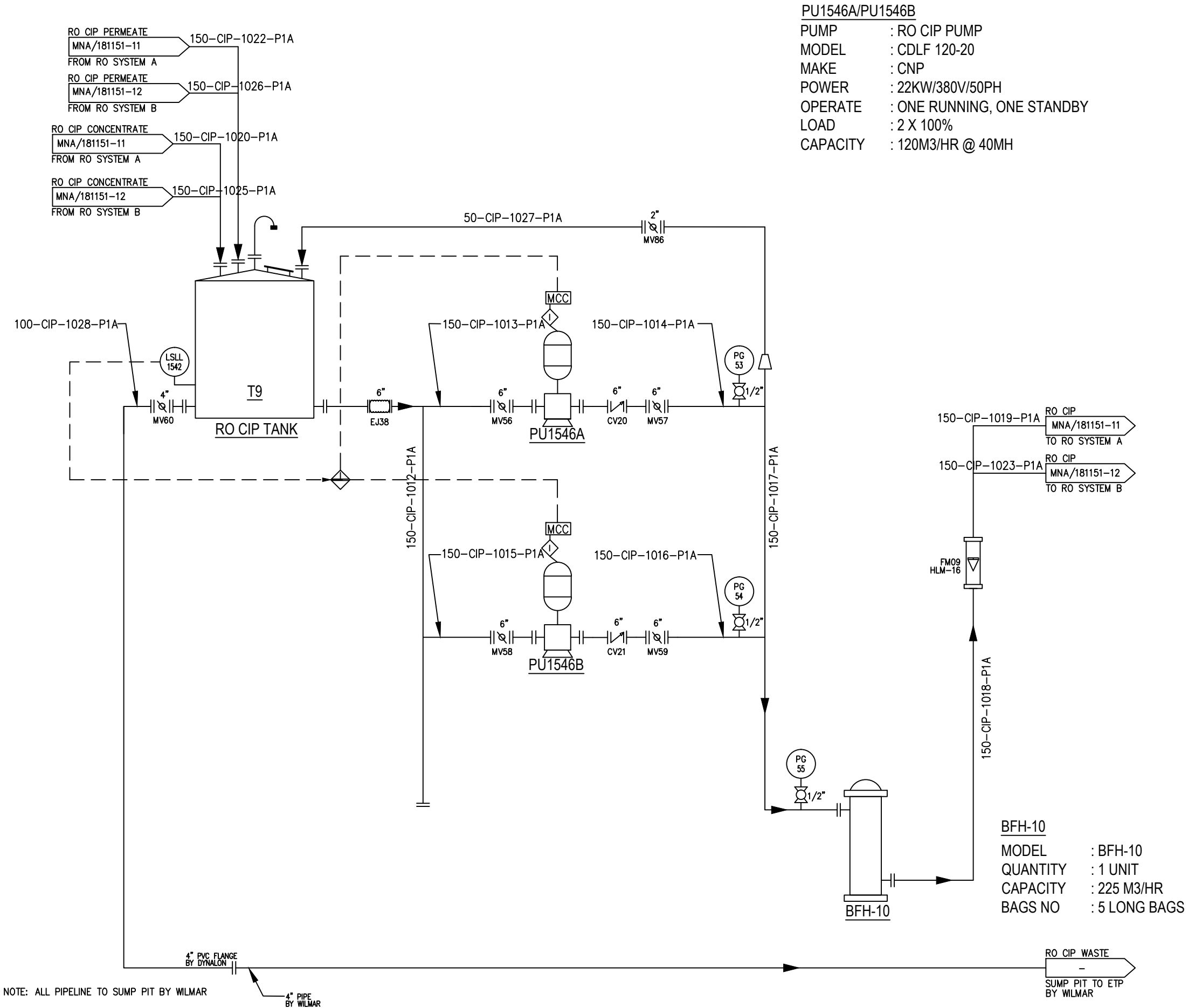
DYNALON ENGINEERING SDN BHD
NO 5, JALAN P2/19A, BANDAR TEKNOLOGI KAJANG,
43500 SEMENYIH SELANGOR, MALAYSIA .
TEL : +603-87236991 / 7991 FAX : +603-87248991

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DRAWN BY	JUNE	SCALE	NTS	
DATE	19.07.2018	DRAWING NO.	MNA/181151-6	REV-17



LEGEND				
REV-17	14/11/2018	CHANGE TAG NO. / RELOCATE FT1535B		
REV-17	14/11/2018	RELOCATE V1533B		
REV	DATE	REMARKS		
OWNER :				
CLIENT :				
WILMAR MNA SERANG				
LOCATION :				
SERANG, INDONESIA				
PROJECT NAME :				
2X200M3/HR AMF, ACP & 2X135M3/HR UF & 2X100M3/HR RO SYSTEM				
DRAWING TITLE :				
PROCESS INSTRUMENTATION DIAGRAM				
CONTRACTOR :				
 DYNALON ENGINEERING SDN BHD NO 5, JALAN P2/19A,BANDAR TEKNOLOGI KAJANG, 43500 SEMENYIH SELANGOR , MALAYSIA . TEL : +603-87236991 / 7991 FAX : +603-87248991				
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DESIGNED BY	CH SHO	APPROVED BY		
DRAWN BY	JUNE	SCALE	NTS	
DATE	19.07.2018	DRAWING NO.	MNA/181151-7	REV-17





PU1546A/PU1546B

PUMP : RO CIP PUMP

MODEL : CDLF 120-20

MAKE : CNP

POWER : 22KW/380V/50PH

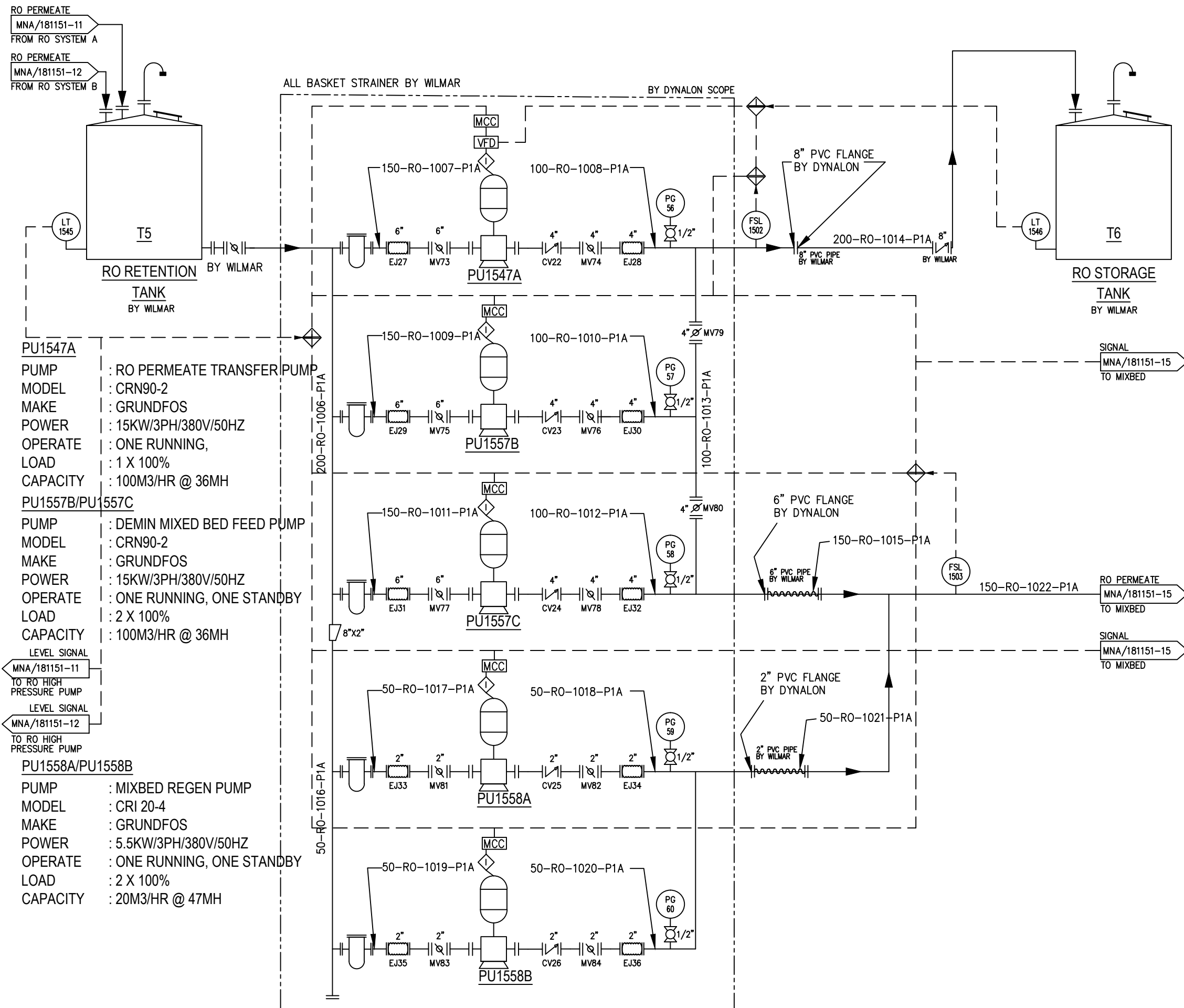
OPERATE : ONE RUNNING, ONE STANDBY

LOAD : 2 X 100%

CAPACITY : 120M3/HR @ 40MH

LEGEND

REV-17	14/11/2018	CHANGE TAG NO.	
REV	DATE	REMARKS	
OWNER :			
CLIENT :			
WILMAR MNA SERANG			
LOCATION :			
SERANG, INDONESIA			
PROJECT NAME :			
2X200M3/HR AMF, ACP & 2X135M3/HR UF & 2X100M3/HR RO SYSTEM			
DRAWING TITLE :			
PROCESS INSTRUMENTATION DIAGRAM			
CONTRACTOR :			
 DYNALON ENGINEERING SDN BHD NO 5, JALAN P2/19A,BANDAR TEKNOLOGI KAJANG, 43500 SEMENYIH SELANGOR , MALAYSIA . TEL : +603-87236991 / 7991 FAX : +603-87248991			
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DATE	19.07.2018	DRAWING NO.	MNA/181151-13
			REV-17



LEGEND

REV	DATE	REMARKS
REV-17	14/11/2018	CHANGE TAG NO.

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
2X135M3/HR UF & 2X100M3/HR
RO SYSTEM

DRAWING TITLE :

PROCESS INSTRUMENTATION DIAGRAM

CONTRACTOR :



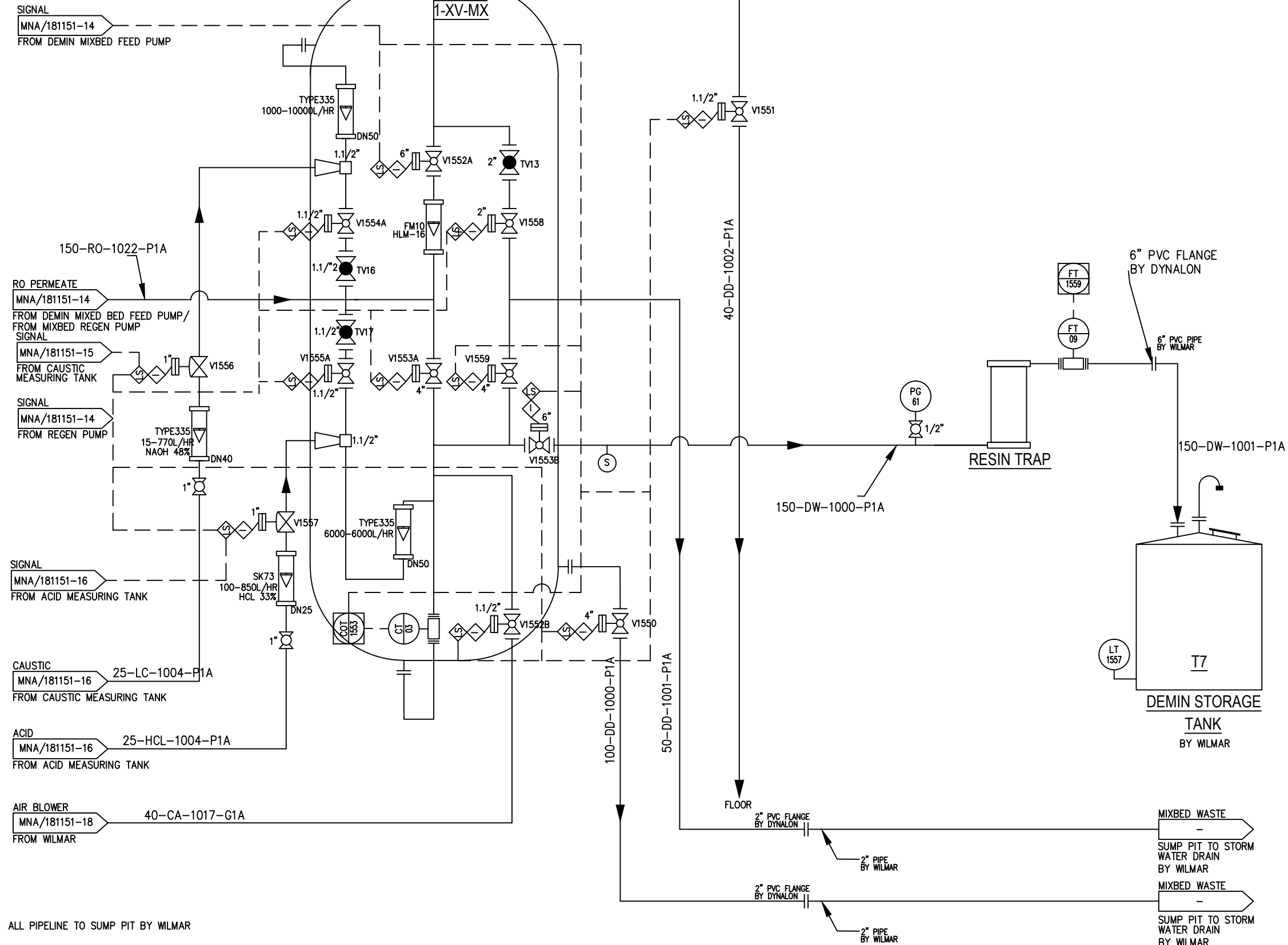
DYNALON ENGINEERING SDN BHD
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DESIGNED BY	CH SHO	APPROVED BY	
DRAWN BY	JUNE	SCALE	NTS
DATE	19.07.2018	DRAWING NO.	MNA/181151-14 REV-17

MIX BED SYSTEM

MODEL : AMD-2500
VESSEL : CARBON STEEL
CAPACITY : 100M3/HR
DIMENSION : 1500MM(DIA) X 3000MM (SH.HT)



LEGEND

REV	DATE	REMARKS
REV-17	14/11/2018	CHANGE TAG NO.

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
2X135M3/HR UF & 2X100M3/HR
RO SYSTEM

DRAWING TITLE :

PROCESS INSTRUMENTATION DIAGRAM

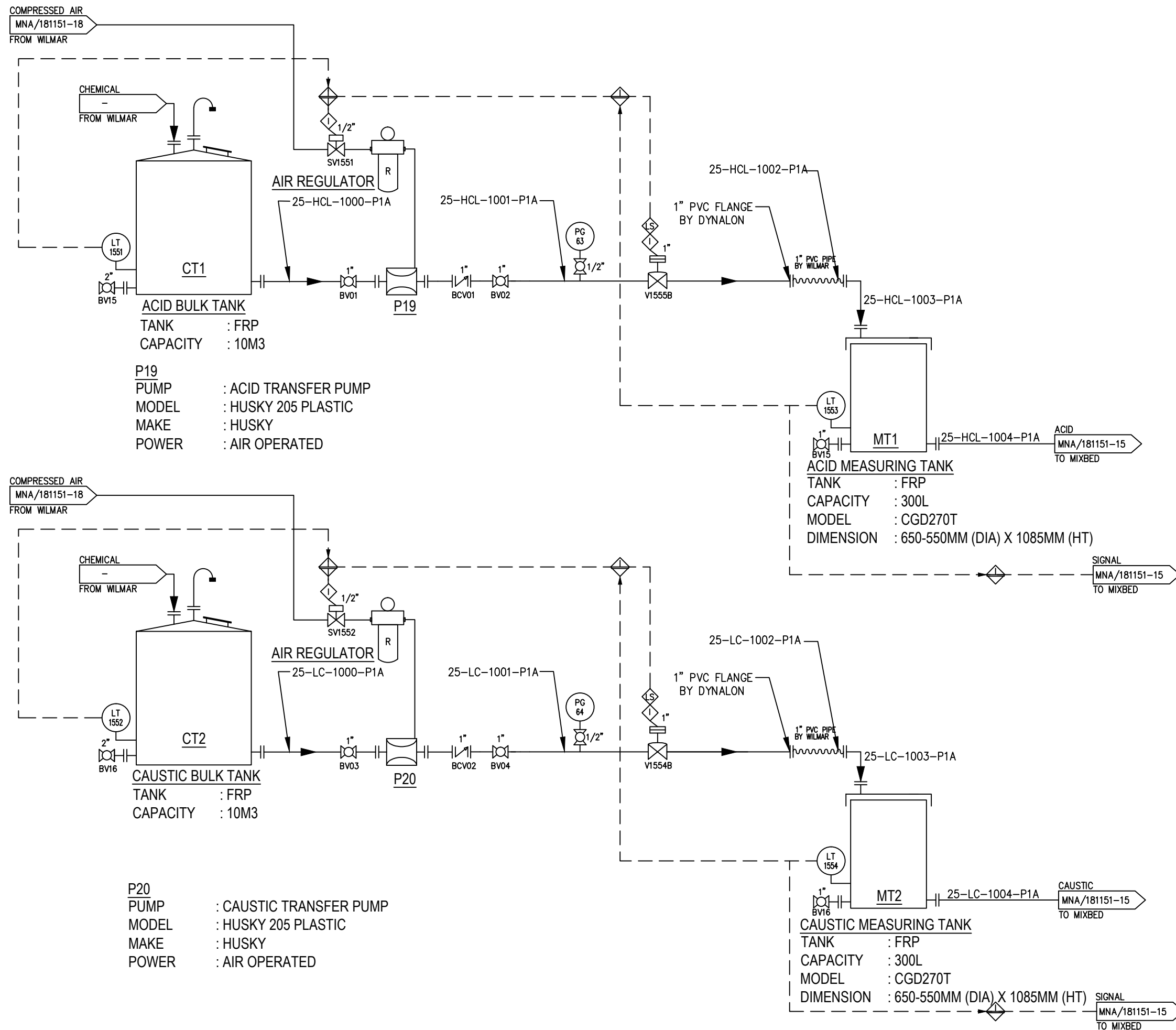
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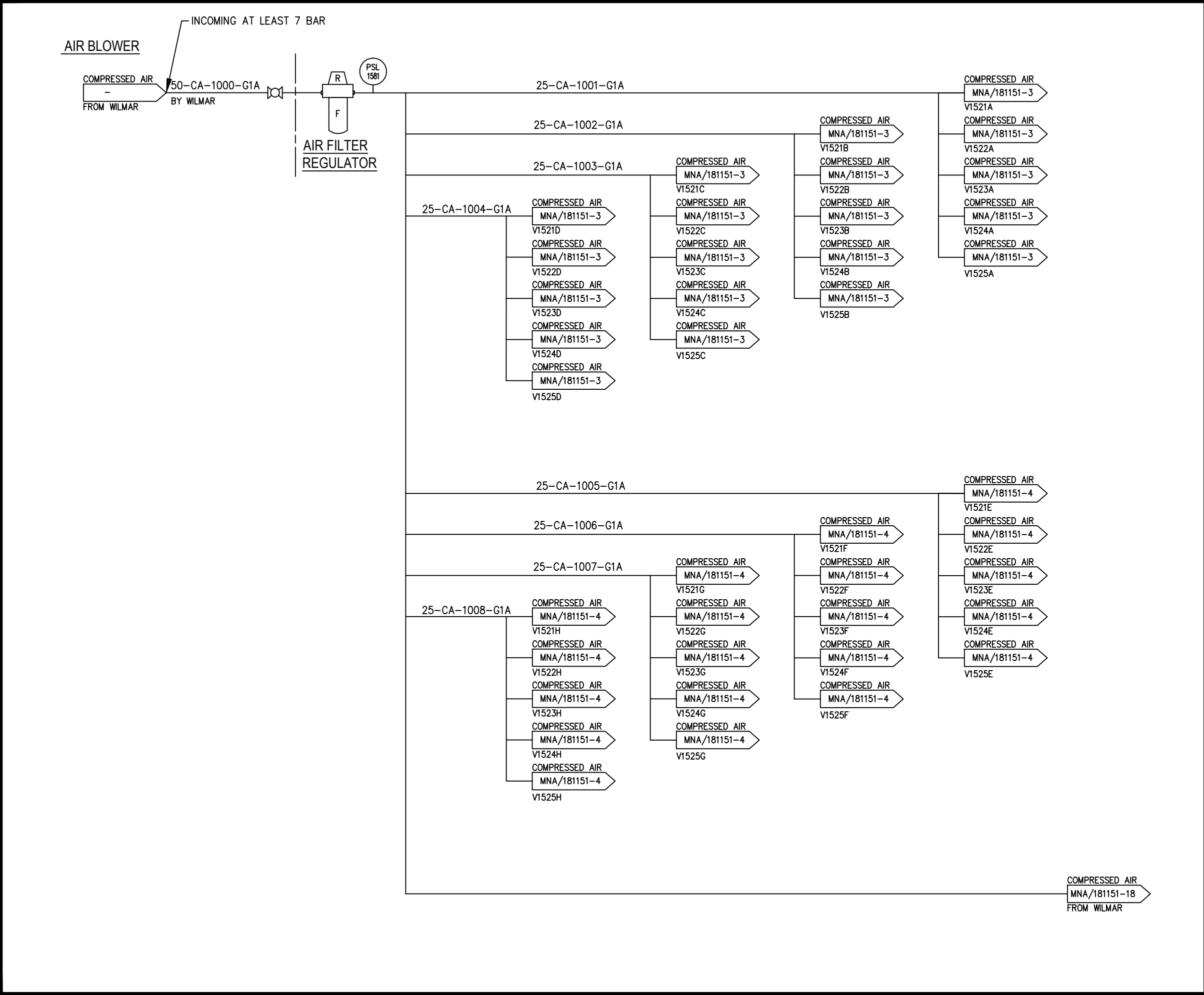


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NO 5, JALAN P2/19A, BANDAR TEKNOLOGI KAJANG,
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DATE	19.07.2018	DRAWING NO.	MNA/181151-15 REV-17

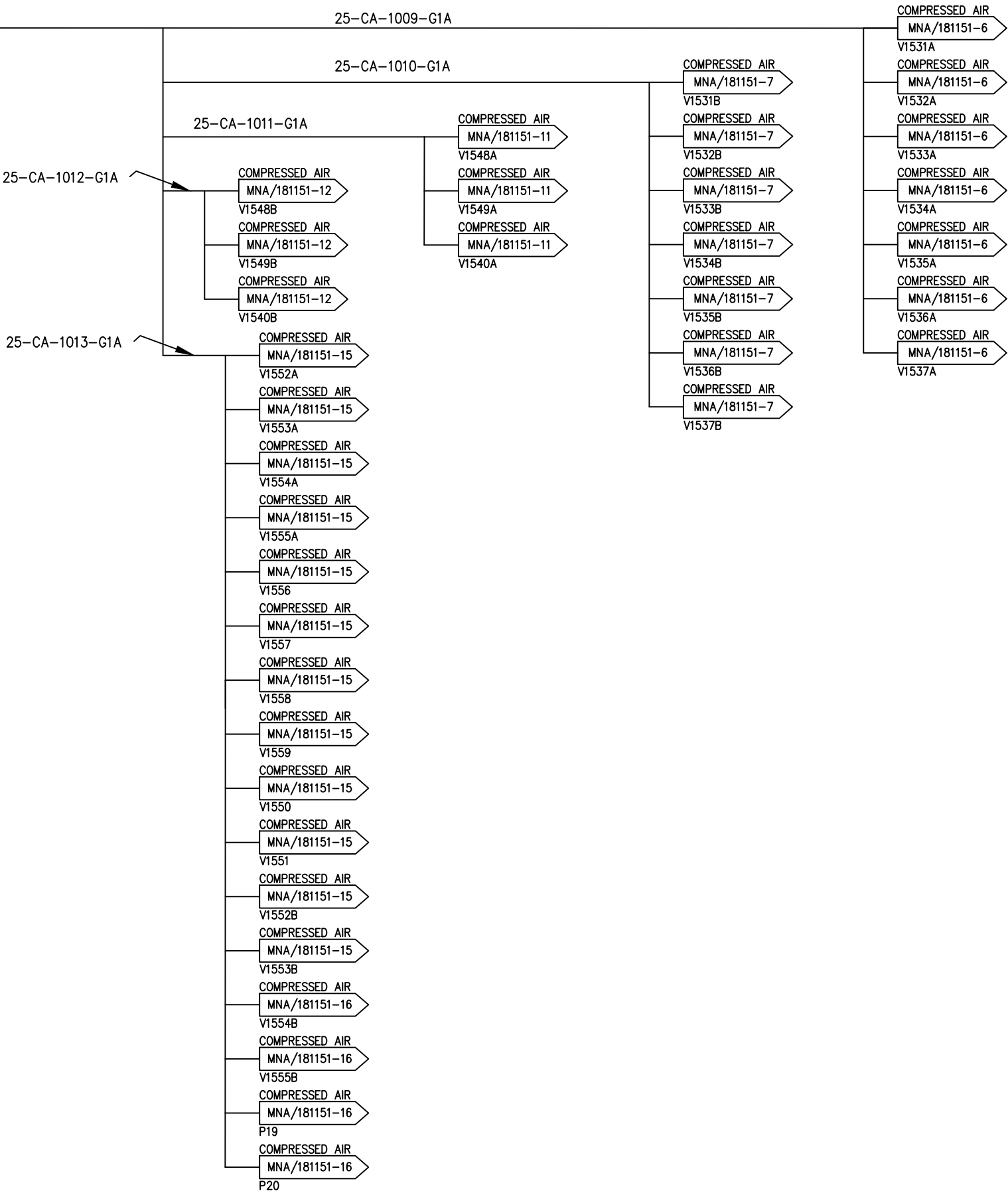




LEGEND				
REV-17	14/11/2018	CHANGE TAG NO.		
REV	DATE	REMARKS		
OWNER :				
CLIENT :				
WILMAR MNA SERANG				
LOCATION :				
SERANG, INDONESIA				
PROJECT NAME :				
2X200M3/HR AMF, ACP & 2X135M3/HR UF & 2X100M3/HR RO SYSTEM				
DRAWING TITLE :				
PROCESS INSTRUMENTATION DIAGRAM				
CONTRACTOR :				
 DYNALON ENGINEERING SDN BHD NO 5, JALAN P2/19A,BANDAR TEKNOLOGI KAJANG, 43500 SEMENYIH SELANGOR , MALAYSIA . TEL : +603-87236991 / 7991 FAX : +603-87248991				
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DRAWN BY	JUNE	SCALE	NTS	
DATE	19.07.2018	DRAWING NO.	MNA/181151-17	REV-17

AIR BLOWER

COMPRESSED AIR
MNA/181151-17
FROM WILMAR



LEGEND

REV-17	14/11/2018	CHANGE TAG NO.
REV	DATE	REMARKS

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
2X135M3/HR UF & 2X100M3/HR
RO SYSTEM

DRAWING TITLE :

PROCESS INSTRUMENTATION DIAGRAM

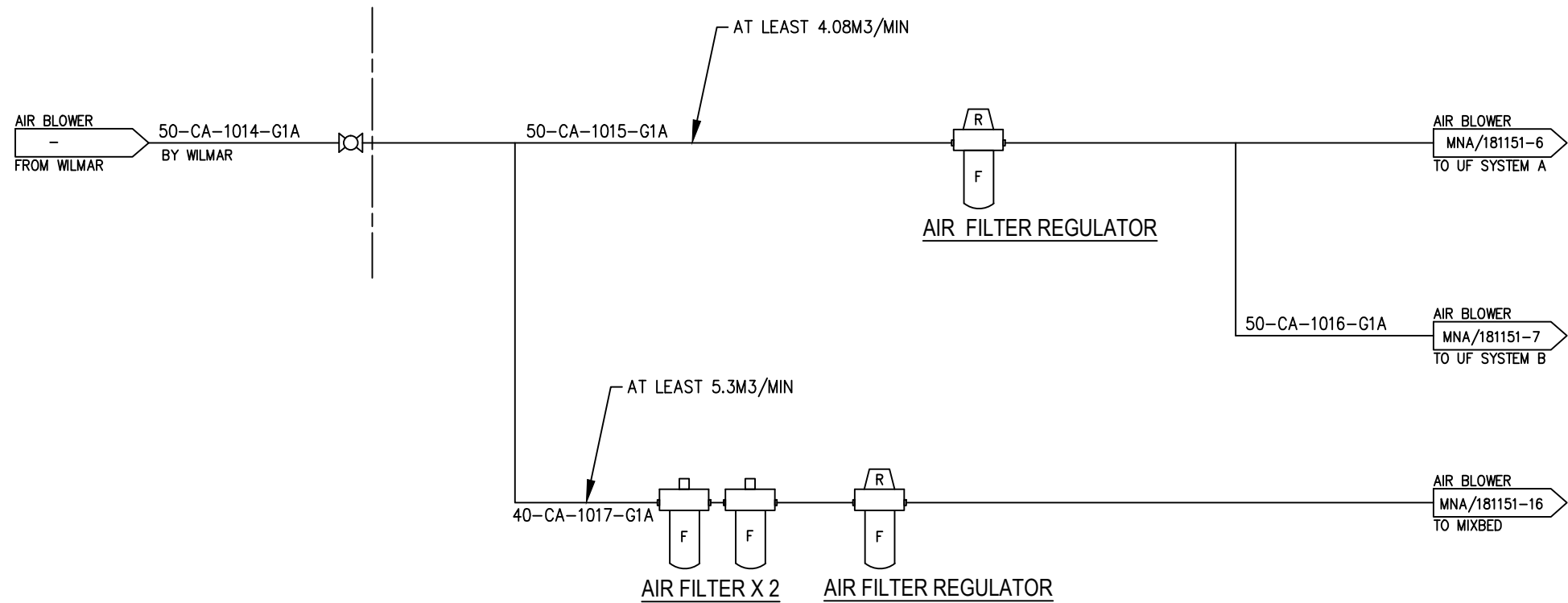
CONTRACTOR :

 DYNALON ENGINEERING SDN BHD
NO 5, JALAN P2/19A, BANDAR TEKNOLOGI KAJANG,
43500 SEMENYIH SELANGOR, MALAYSIA.
TEL : +603-87236991 / 7991 FAX : +603-87248991

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DATE	19.07.2018	DRAWING NO.	MNA/181151-18 REV-17

AIR BLOWER



LEGEND

REV-17 14/11/2018 CHANGE TAG NO., REMOVE FILTER

REV DATE REMARKS

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
2X135M3/HR UF & 2X100M3/HR
RO SYSTEM

DRAWING TITLE :

PROCESS INSTRUMENTATION DIAGRAM

CONTRACTOR :

 DYNALON ENGINEERING SDN BHD
NO 5, JALAN P2/19A,BANDAR TEKNOLOGI KAJANG,
43500 SEMENYIH SELANGOR , MALAYSIA .
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DESIGNED BY	CH SHO	APPROVED BY		
DRAWN BY	JUNE	SCALE	NTS	
DATE	19.07.2018	DRAWING NO.	MNA/181151-19	REV-17

VALVE SYMBOLS

	Ball Valve
	Globe Valve
	Gate Valve
	Check Valve
	Needle Valve
	Butterfly Valve
	Plug Valve
	Knife Valve
	Diaphragm Valve
	Ball Float Valve
	Actuator Valve c/w Namur Valve & Limit Switch
	Control Valve
	Sampling Valve
	Solenoid Valve

SELF-ACTUATED DEVICES

	Pressure Relief or Safety Valve
	Vacuum Relief Valve
	Breather Valve
	Pressure Reducing Regulator (Self Contained)
	Back Pressure Regulator (Self Contained)
	Pressure Reducing Regulator W/ External Tap
	Back Pressure Regulator W/ External Tap

STEAM UTILITIES SYMBOLS

	Ball Float Steam Trap
	Steam Separator
	Thermostatic Steam Trap

PIPING SPECIALITY ITEMS

	Y-Type Strainer
	Basket Type Strainer
	T-Type Strainer
	Oriface
	Expantion Joint
	Flexible Hose
	Sight Glass
	Silencer

PIPING FITTINGS

	Concentric Reducer
	Eccentric Reducer
	Cap
	Flange
	Blind Flange
	Hose Connection

INSTRUMENT LINE SYMBOLS

	Pneumatic Signal
	Electric Signal
	Connection to Process Or Instrument Supply
	Mechanical Link
	Software or Data Link
	Signal Line

PIPE LINE COLOR & SYMBOLS

	Process Water Line
	Olein Line
	Air Compressor Line
	Cooling Water Line
	Chiler Water Line
	Low/Medium/High Pressure Steam Line
	Vessel Insulation
	Heat Insulation & Tracing Line
	Insulation Line

LINE DESCRIPTION

XX-XXX-XXXX-XX-XXXX	Thickness Insulation (mm)
	Insulation Type
	Corrosion Allowance
	Presure Rating
	Material
	Line Number
	Service / Media Code
	Line Size (DN)

PIPING SPECIFICATION

Material
C : Carbon Steel A53/A106 B
D : Duplex SS
R : Stainless Steel 316/316L
S : Stainless Steel 304/304L
G : CS A53/A106 B Galvanized
P : PVC SCH80
H : HDPE

Pressure Rating

1 - ANSI Class 150
2 - ANSI Class 300
3 - ANSI Class 600
12 - DIN PN 16
13 - DIN PN 25
14 - DIN PN 40

Corrosion Allowance

A - 0.0mm, ASME B31.3
B - 1.6mm, ASME B31.3
C - 3.2mm, ASME B31.3

Insulation Type

RW - Rock Wool
PU - Poliurethane
ST - Steam Tracing

EQUIPMENT SYMBOLS

	Centrifugal Pump
	Gear Pump
	Diafragma Pump
	Vertical Pump
	Vacumm Pump
	Screw Pump
	Reciprocating Pump
	Plate Heat Exchanger
	Spiral Heat Exchanger
	Shell & Tube Heat Exchanger
	Air Compressor
	Ejector

GENERAL INSTRUMENT SYMBOLS

LOCATION/ACCESSIBILITY	DISCRETE INSTRUMENT	SHARED DISPLAY & CONTROL (DCS)	PLC
Field Mounted : 1. Field or locally mounted 2. Accessible to an operator at device			
Primary Location Normally Accessible to an Operator 1. Central or main control room 2. Front or main panel or console mounted 3. Visible on video display 4. Accessible to an operator at device or console			
Primary Location Normally Inaccessible to an Operator 1. Central or main control room 2. Rear of panel or cabinet mounted 3. Not visible on video display 4. Not normally accessible to an Operator at device or console			

MEDIA CODE

HCL	: Hydrocloric Acid
PA	: Phosphoric Acid
CAC	: Activated Carbon
LC	: Sodium Hydroxide
SM	: Sodium Methilate
N2	: Nitrogen
TOH	: Thermal Oil
LPC	: Low Pressure Condensate
MPC	: Medium Pressure Condensate
HPC	: High Pressure Condensate
HPS	: High Pressure Steam
MPS	: Medium Pressure Steam
LPS	: Low Pressure Steam
NG	: Natural Gas
IA	: Instrument Air
CA	: Compressed Air / Utility Air
V	: Vacuum
W	: City / Process Water
BFW	: Boiler Feed Water
CW	: Cooling Water
COL	: Crude Alcohol
TW	: Tempered Cooling Water
DW	: Demineralized Water
RO	: Reverse Osmosis Water
WW	: Waste Water
CPO	: Crude Palm Oil
PFAD	: Fatty Acid Destilate Oil
RPO	: Refined Palm Oil
ROL	: Refined Olein
RPS	: Refined Palm Stearin
PME	: Palm Methyl Esther
DSL	: Diesel Oil
FW	: Filtered Water
UF	: Ultrafiltration Water
RC	: RO Concentrate Water
CIP	: Membrane Cleaning Chemical
PAC	: Polyaluminium Chloride
SH	: Sodium Hypochlorite

LEGEND

REV	DATE	REMARKS

OWNER :

CLIENT :

WILMAR MNA SERANG

LOCATION :

SERANG, INDONESIA

PROJECT NAME :

2X200M3/HR AMF, ACP &
2X135M3/HR UF & 2X100M3/HR
RO SYSTEM

DRAWING TITLE :

SYMBOL & LEGEND OF P&ID

CONTRACTOR :

	DYNALON ENGINEERING SDN BHD NO 5, JALAN P2/19A,BANDAR TEKNOLOGI KAJANG, 43500 SEMENYIH SELANGOR , MALAYSIA . TEL : +603-87236991 / 7991 FAX : +603-87248991
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DESIGNED BY	CH SHO	APPROVED BY	
DRAWN BY	JUNE	SCALE	NTS
DATE	19.07.2018	DRAWING NO.	MNA/181151-20 REV-01